Explore the evidence

Explore combustion research. See what was observed—and where the evidence stops.

Scenario

Preset

Illustrative lunar rover ˅

ISS acrylic / Illustrative lunar rover / Custom

Research question

What did NASA observe when airflow changed around burning acrylic on the ISS, and which parts carry over to an illustrative lunar rover?

Task

Materials & ventilation ˅

Material

PMMA / Acrylic ˅

Geometry

Rod ˅

Characteristic length

9.525 mm

Gravity

Lunar 1/6 g ˅

Oxygen

34 %

Pressure

56.5 kPa / 8.2 psia

Airflow

Not specified · cm/s

Diluent gas

N_2 ˅

Requested outcome

Spread rate ˅

Inferred from question: requested outcome. Review the resolved scenario before interpreting results.

Evaluate scenario

Copy scenario link

Shareable scenario settings only. Question text, secrets, and prompts are not retained in the URL.

Reset scenario

No verified run tests all requested conditions together

Individual parameters appear across separate studies, but no single verified run tests all requested conditions and outcome jointly.

PMMA rod · 9.525 mm · lunar 1/6 g · 34% O_2 · 56.5 kPa · N_2 · spread rate

Relevant investigations

Study relevance—not a safety rating. Scores are computed from the resolved scenario.

BASS-II 81.9 / 100

Verified candidate run data · 2 reviewed runs · 24 published runs

Best match for materials and ventilation on PMMA rods; provides ISS microgravity forced-flow data. Lunar 1/6 g and 56.5 kPa remain untested in BASS-II.

View evidence → · Relevance breakdown below

Why this study ranks here

Category	Base / normalized weight	Match / points	Status
Research objective	35 / 35	1.0 / 35	Matched

Reason: Investigation directly targets materials_and_ventilation.

Omitted fields have their weights renormalized. Unknown fields earn 0 points and remain explicitly marked unknown.

Show all relevance categories ↓

Also in the collection: Saffire-I · FLEX · SAME-R. Ordering and additional scores follow the evaluation response.

LUCI · published reference only

1 published run · 0 reviewed runs · New Shepard sounding rocket

A rotating payload produced ~0.16 g. A 4 mm diameter (2 mm radius) PMMA rod and cotton-fiberglass fabric burned simultaneously in normal-pressure air: 21% O_2 and 101.3 kPa.

Shared-chamber oxygen is not a single-sample measurement

The rod and fabric burned inside one chamber. Oxygen sensor readings reflect their shared consumption and cannot be attributed to either sample alone.

Inspect EV_LUCI_OVERVIEW · EV_LUCI_SHARED_O2 →

Where the evidence applies

Joint coverage: not tested · Matching verified runs: none

Requested parameter	Coverage	Evidence
PMMA	Tested in catalog	O15a / O15b · BASS-II; LUCI
Lunar 1/6 g	Published reference only	LUCI · no reviewed run
34% oxygen	Not tested in catalog	No matching runs
56.5 kPa (8.2 psia)	Not tested in catalog	No matching runs

A match on one parameter is not proof that the full scenario was tested.

Reviewed runs

2 selected · aliases O15a / O15b resolve to canonical run identifiers.

Selected run	Initial O_2	Initial opposing flow	Protocol
✓ BASSII_O15A	17.5%	10 cm/s	Changes during test
✓ BASSII_O15B	17.5%	9 cm/s	Changes during test

Shared clear PMMA sample 218m · 9.525 mm diameter. Run-level pressure is unreported.

Compare 2 selected runs

What the evidence says

Reviewed fallback · provider not configured

Source-bound synthesis remains available when AI is unavailable, slow, rate-limited, or fails citation validation.

Observed findings

O15a and O15b used shared sample 218m, a clear 9.525 mm PMMA rod. Tests started at 17.5% oxygen and 10 cm/s and 9 cm/s initial opposing flow; flow changed during each test.

EV_O15A_TABLE_A5 · EV_O15B_TABLE_A5 →

Applicability limits

The illustrative lunar rover scenario—lunar 1/6 g, 34% oxygen, and 56.5 kPa / 8.2 psia—has no jointly tested verified run in the catalog.

EV_LUCI_OVERVIEW · EV_O15A_TABLE_A5 →

Missing evidence

Run-level pressure for O15a and O15b is unreported in Table A.5. Intermediate elapsed times and flow velocities in Figure 6.4 are unreported.

EV_O15A_PRESSURE_UNREPORTED · EV_FIG64_PHASES →

Research observations are not cabin safety limits

Figure 6.4 reports quenching at 13.6% oxygen in ISS microgravity on a 6.35 mm black PMMA rod—not certified spacecraft cabin safety thresholds or operational flammability limits. Do not infer a safe cabin oxygen setting from these records.

Compare reviewed runs

Run A: BASSII_O15A — Verified run

Run B: BASSII_O15B — Verified run

Interpret with care: shared sample 218m and changing airflow—not an isolated airflow-only comparison.

Caveats below ↓

Parameter	O15a	O15b	
SAMPLE & APPARATUS			
Material	Clear cast PMMA	Clear cast PMMA	
Geometry	Rod	Rod	
Physical sample	218m	218m	
Rod diameter	9.525 mm	9.525 mm	
Platform	ISS	ISS	
EXPERIMENTAL CONDITIONS			
Gravity	Microgravity	Microgravity	
Flow direction	Opposing (initial)	Opposing (initial)	
Flow protocol	Changing during test	Changing during test	
Pressure	*Not reported*	*Not reported*	Unknown
MEASURED QUANTITIES			
Initial airflow	10 cm/s	9 cm/s	Different
Initial oxygen	17.5%	17.5%	Same
Measurement basis	Initial; flow changes	Initial; flow changes	Same protocol
INTERPRETATION			
Cabin safety threshold	Not established	Not established	

Missing values remain unknown; nominal cabin pressure is not substituted.

EVIDENCE SOURCE
NASA/TM-20210011385
Table A.5 · EV_O15A_TABLE_A5
Inspect evidence →

EVIDENCE SOURCE
NASA/TM-20210011385
Table A.5 · EV_O15B_TABLE_A5
Inspect evidence →

Catalog 2026.11.14-66a836f1 · Values aligned with the integration handoff

What this comparison can—and cannot—show

Both reviewed runs use the same clear 9.525 mm PMMA rod, sample 218m, in ISS microgravity. Initial oxygen is 17.5% in both runs. Initial opposing airflow is 10 cm/s for O15a and 9 cm/s for O15b; airflow changes during each test.

Unknown is not zero: run-level ambient pressure is unreported in Table A.5. Nominal ISS cabin pressure is never substituted.

Keep Figure 6.4 separate: its four phases are a sequence on a 6.35 mm black PMMA rod, not four additional runs and not the clear 9.525 mm sample 218m.

Evidence gap: these are research observations, not certified cabin oxygen limits or operational flammability thresholds.

Sources & evidence

9 sources · 12 evidence records

Search sources by title, identifier, keyword...

- **BASS-II technical report** — NASA/TM-20210011385
- **PMMA concurrent-flow cohort** — NTRS-20150008961
- **BASS-II dataset** — PSI-25
- **Saffire-I dataset** — PSI-98
- **SAME-R smoke aerosol dataset** — PSI-101
- **FLEX droplet dataset** — PSI-69
- **LUCI** — NTRS-20250010653
- **Exploration atmosphere** — NTRS-20240003211
- **Partial-gravity review** — NTRS-20220015800

O15a · Initial oxygen

Individual run · Author-reported

Reviewed fact

Clear 9.525 mm PMMA rod, sample 218m. O15a started at 17.5% oxygen and 10 cm/s opposing flow; airflow changed during the test. Run-level pressure is unreported.

Original **17.5 % O_2** / Normalized **17.5 % · Initial**

Source	NASA/TM-20210011385
Printed page	110
PDF page	122
Locator	Appendix A · Table A.5 · O15a
Scope	Individual run
Measurement basis	Initial

Open original source

A source citation is a starting point for review.

EV_O15A_TABLE_A5

Catalog 2026.11.14-66a836f1 · Source-bound evidence

Trace each statement to its evidence

O15a / O15b · EV_O15A_TABLE_A5 / EV_O15B_TABLE_A5 · SRC_BASSII_TM2021

Unreported pressure · EV_O15A_PRESSURE_UNREPORTED · SRC_BASSII_TM2021

Figure 6.4 sequence · EV_FIG64_PHASES · SRC_BASSII_TM2021

LUCI reference · EV_LUCI_OVERVIEW / EV_LUCI_SHARED_O2 · SRC_LUCI_NTRS2025

Only evidence and sources associated with this evaluation may support its explanation. A citation from another study does not establish the requested combination of conditions.

About this collection

What is available, and where the evidence stops.

5	9	2	4	12	3
Investigations	Sources	Verified runs	Phases	Evidence records	Review records

Studies, runs, and phases

Five investigation profiles. Two individually reviewed runs.

BASS-II	2 reviewed runs
Saffire-I	Study summary
SAME-R	Study summary
FLEX	Published reference
LUCI	Published reference

24 published cohort runs; 2 reviewed candidates; remaining 22 require review.

Download the records

Download curated records and catalog resources.

Verified experiment runs Individually reviewed experiment rows (n = 2).	CSV ↓
Phase observations Phase-level observations (n = 4).	CSV ↓
Data dictionary Field definitions, units, and value conventions.	JSON ↓
Catalog manifest This catalog's contents, provenance, and scope.	JSON ↓

Review log

Remaining 22 PMMA cohort rows Pending review

Figure 6.4 intermediate measurements — Unreported
Intermediate times and flow velocities remain unknown; never interpolate.

LUCI run-level telemetry — Not an isolated reviewed run
Published shared-chamber evidence is available as a reference, not isolated-run telemetry.

Catalog identity

Version 2026.11.14-66a836f1

Catalog snapshot, not a live refresh timestamp.

A sequence, not four separate runs

Stage 1	Stage 2	Stage 3	Stage 4
Initial	Intermediate	Intermediate	Final
14.8% O_2	Not reported	Not reported	13.6% O_2

Figure 6.4 · 6.35 mm black PMMA rod · ordered phases, not elapsed time. Intermediate times and ve Evidence reported.

Catalog integrity & export metadata

Version · 2026.11.14-66a836f1

SHA-256 · 66a836f15a94c518f0a4bc7213e4710bfbe772e2b54fd3183d08276511ff21bb

Runs CSV · 2 reviewed rows / Phases CSV · 4 ordered observations

Data dictionary · canonical units, tolerances, and schema documentation

Manifest · catalog version, full integrity hash, counts, and per-export SHA-256 digests

CSV downloads retain their data-version and row-count metadata.

Release status — three separate axes

Software readiness · Complete

The supplied handoff reports the evaluation engine, comparison, coverage matrix, citation validation, and 15 tests passing.

Scientific review · Limited reviewed scope

2 verified candidate runs + 4 verified phases + 5 investigation profiles. Additional PMMA cohort data and PSI telemetry require access and human reviewer sign-off.

Live provider · Optional; not live-verified by this snapshot

Simulated provider adapters are verified. A live provider requires runtime configuration; reviewed fallback remains available without it.

Explore the evidence

Explore combustion research. See what was observed—and where the evidence stops.

Scenario

Preset

Illustrative lunar rover

ISS acrylic / Illustrative lunar rover / Custom

Research question

What did NASA observe when airflow changed around burning acrylic on the ISS, and which parts carry over to an illustrative lunar rover?

Task

Materials & ventilation

Material

PMMA / Acrylic

Geometry

Rod

Characteristic length

9.525 mm

Gravity

Lunar 1/6 g

Oxygen

34 %

Pressure

56.5 kPa / 8.2 psia

Airflow

Not specified · cm/s

Diluent gas

N_2

Requested outcome

Spread rate

Inferred from question: requested outcome. Review the resolved scenario before interpreting results.

Evaluate scenario

Copy scenario link

Shareable scenario settings only. Question text, secrets, and prompts are not retained in the URL.

Reset scenario

No verified run tests all requested conditions together

Individual parameters appear across separate studies, but no single verified run tests all requested conditions and outcome jointly.

PMMA rod · 9.525 mm · lunar 1/6 g · 34% O_2 · 56.5 kPa · N_2 · spread rate

Relevant investigations

Study relevance—not a safety rating. Scores are computed from the resolved scenario.

BASS-II
81.9 / 100

Verified candidate run data · 2 reviewed runs · 24 published runs

Best match for materials and ventilation on PMMA rods; provides ISS microgravity forced-flow data. Lunar 1/6 g and 56.5 kPa remain untested in BASS-II.

View evidence → · Relevance breakdown below

Why this study ranks here

Category: Research objective

Base / normalized weight: 35 / 35

Match / points: 1.0 / 35

Status: Matched

Reason: Investigation directly targets materials_and_ventilation.

Omitted fields have their weights renormalized. Unknown fields earn 0 points and remain explicitly marked unknown.

Show all relevance categories ↓

Also in the collection: Saffire-I · FLEX · SAME-R. Ordering and additional scores follow the evaluation response.

LUCI · published reference only

1 published run · 0 reviewed runs · New Shepard sounding rocket

A rotating payload produced ~0.16 g. A 4 mm diameter (2 mm radius) PMMA rod and cotton-fiberglass fabric burned simultaneously in normal-pressure air; 21% O_2 and 101.3 kPa.

Shared-chamber oxygen is not a single-sample measurement

The rod and fabric burned inside one chamber. Oxygen sensor readings reflect their shared consumption and cannot be attributed to either sample alone.

Inspect EV_LUCI_OVERVIEW · EV_LUCI_SHARED_02 →

Where the evidence applies

Joint coverage: not tested · Matching verified runs: none

Requested parameter: PMMA
Coverage: Tested in catalog
Evidence: O15a / O15b · BASS-II; LUCI

Requested parameter: Lunar 1/6 g
Coverage: Published reference only
Evidence: LUCI · no reviewed run

Requested parameter: 34% oxygen
Coverage: Not tested in catalog
Evidence: No matching runs

Requested parameter: 56.5 kPa (8.2 psia)
Coverage: Not tested in catalog
Evidence: No matching runs

A match on one parameter is not proof that the full scenario was tested.

Reviewed runs

2 selected · aliases O15a / O15b resolve to canonical run identifiers.

Selected run: ✓ BASSII_O15A
Initial O_2: 17.5%
Initial opposing flow: 10 cm/s
Protocol: Changes during test

Selected run: ✓ BASSII_O15B
Initial O_2: 17.5%
Initial opposing flow: 9 cm/s
Protocol: Changes during test

Shared clear PMMA sample 218m · 9.525 mm diameter. Run-level pressure is unreported.

Compare 2 selected runs

What the evidence says

Reviewed fallback · provider not configured

Source-bound synthesis remains available when AI is unavailable, slow, rate-limited, or fails citation validation.

Observed findings

O15a and O15b used shared sample 218m, a clear 9.525 mm PMMA rod. Tests started at 17.5% oxygen and 10 cm/s and 9 cm/s initial opposing flow; flow changed during each test.

EV_O15A_TABLE_A5 · EV_O15B_TABLE_A5 →

Applicability limits

The illustrative lunar rover scenario—lunar 1/6 g, 34% oxygen, and 56.5 kPa / 8.2 psia—has no jointly tested verified run in the catalog.

EV_LUCI_OVERVIEW · EV_O15A_TABLE_A5 →

Missing evidence

Run-level pressure for O15a and O15b is unreported in Table A.5. Intermediate elapsed times and flow velocities in Figure 6.4 are unreported.

EV_O15A_PRESSURE_UNREPORTED · EV_FIG64_PHASES →

Research observations are not cabin safety limits

Figure 6.4 reports quenching at 13.6% oxygen in ISS microgravity on a 6.35 mm thick PMMA rod—not certified spacecraft cabin safety thresholds or operational flammability limits. Do not infer a safe cabin oxygen setting from these records.

Compare reviewed runs

Two reviewed candidates. Same sample; changing airflow.

Run O15a ⌄

BASSII_O15A

Material · Clear PMMA

Geometry · Rod

Physical sample · 218m

Diameter · 9.525 mm

Platform · ISS microgravity

Initial oxygen · 17.5%

Initial opposing airflow · 10 cm/s

Flow protocol · Changes during test

Run-level pressure · Not reported

Cabin safety threshold · Not established

Inspect EV_O15A_TABLE_A5 →

Run O15b ⌄

BASSII_O15B

Material · Clear PMMA

Geometry · Rod

Physical sample · 218m

Diameter · 9.525 mm

Platform · ISS microgravity

Initial oxygen · 17.5%

Initial opposing airflow · 9 cm/s

Flow protocol · Changes during test

Run-level pressure · Not reported

Cabin safety threshold · Not established

Inspect EV_O15B_TABLE_A5 →

What differs

Initial opposing airflow: O15a 10 cm/s; O15b 9 cm/s.

Initial oxygen is the same: 17.5%. Both records use sample 218m, but flow changes within each run. Do not interpret this as an isolated airflow-only comparison.

Missing is not zero

Run-level pressure remains unknown. Nominal ISS cabin pressure is not substituted.

Figure 6.4 is a separate four-phase sequence on a 6.35 mm black PMMA rod—not four runs and not sample 218m.

These research observations do not establish safe cabin oxygen settings.

Source · NASA/TM-20210011385

Sources & evidence

9 sources · 12 evidence records

Search sources

Title, identifier, keyword...

Research sources

BASS-II technical report ↗

NASA/TM-20210011385

PMMA concurrent-flow cohort ↗

NTRS-20150008961

BASS-II dataset ↗

PSI-25

Saffire-I dataset ↗

PSI-98

SAME-R smoke aerosol dataset ↗

PSI-101

FLEX droplet dataset ↗

PSI-69

LUCI ↗

NTRS-20250010653

Exploration atmosphere ↗

NTRS-20240003211

Partial-gravity review ↗

NTRS-20220015800

O15a · Initial oxygen

Individual run · Author-reported

Clear 9.525 mm PMMA rod, sample 218m. O15a started at 17.5% oxygen and 10 cm/s opposing flow. Airflow changed during the test; run-level pressure is unreported.

Original

17.5 % O_2

Normalized

17.5 % · Initial

Source

NASA/TM-20210011385

Printed / PDF page

110 / 122

Locator

Appendix A · Table A.5 · O15a

Scope

Individual run

Measurement basis

Initial

Evidence ID

EV_O15A_TABLE_A5

Open original source ↗

Close evidence

Related evidence

EV_O15B_TABLE_A5 →

EV_O15A_PRESSURE_UNREPORTED →

EV_FIG64_PHASES →

EV_LUCI_OVERVIEW →

EV_LUCI_SHARED_O2 →

Only sources and evidence associated with this evaluation support its explanation. A citation alone does not establish the requested joint conditions.

About this collection

What is available—and where the evidence stops.

Collection at a glance

5 investigations · 9 sources

2 verified runs · 4 phases

12 evidence records · 3 review records

Studies, runs, and phases

BASS-II · 2 reviewed candidate runs

Saffire-I · Study summary

SAME-R · Study summary

FLEX · Published reference

LUCI · Published reference

24 published PMMA cohort runs does not mean 24 reviewed experiments. The remaining 22 require review.

Download the records

Verified runs · CSV ↓

2 individually reviewed experiment rows

Phase observations · CSV ↓

4 ordered observations from Figure 6.4

Data dictionary · JSON ↓

Canonical units, tolerances, and schema

Catalog manifest · JSON ↓

Version, counts, and SHA-256 digests

CSV metadata retains data version and row count.

Review log

Remaining 22 PMMA cohort rows · Pending review

PSI telemetry requires access and human reviewer sign-off.

Figure 6.4 intermediate measurements · Unreported

Intermediate elapsed times and flow velocities remain unknown. Never interpolate them.

LUCI telemetry · Not an isolated reviewed run

Shared-chamber observations remain published reference evidence.

A sequence—not four runs

Figure 6.4 · 6.35 mm black PMMA rod

Stage 1 · Initial · 14.8% O_2

Stage 2 · Intermediate · Not reported

Stage 3 · Intermediate · Not reported

Stage 4 · Final · 13.6% O_2

Stage order is not elapsed time. Intermediate times and velocities are unreported.

Inspect EV_FIG64_PHASES →

Catalog identity & integrity

2026.11.14-66a836f1

Snapshot version, not a live refresh timestamp.

SHA-256

66a836f15a94c518f0a4bc7213e4710bfbe772e2b54fd3183d08276511ff21bb

Use the manifest to inspect catalog counts and per-export digests.

Release status

Software · Complete in supplied handoff

Evaluation, comparison, coverage, citation validation, and 15 tests reported passing.

Scientific review · Limited scope

2 verified candidates + 4 verified phases + 5 investigations. Further data requires review.

Live provider · Optional

Simulated adapter verified. Runtime configuration is required for live-provider use; reviewed fallback works without it.

COMBUSTION RESEARCH / SPACE SCIENCE

Explore fire research in space.

A clearer way to explore combustion experiments beyond Earth.

Find relevant studies, compare reviewed experiments, and understand exactly where the evidence applies.

What would you like to investigate?

ISS acrylic experiment | Lunar rover scenario | Build your own

SCROLL TO EXPLORE

Evidence first.
Every claim has a source.

01 / THE PLATFORM

From a question to evidence you can trust.

Flame Fall brings scattered space-combustion research into one focused workspace — without confusing a related study with a verified match.

01 / DISCOVER

Explore the research

Search investigations and narrow results by material, geometry, gravity, and conditions.

02 / COMPARE

Compare experiments

Review runs side by side. See what matches, differs, and was never reported.

03 / VERIFY

Trace every claim

Follow findings back to source documents, page references, and measured evidence.

BUILT AROUND THE EVIDENCE

Not more noise. Better scientific context.

Workspace states & integration

Companion states for the research workspace · same scenario, separate evaluation and explanation lifecycles.

These are design specifications, not a claim that services are currently connected.

01 / Evaluation ready · explanation loading

BASS-II 81.9 / 100 · **2 reviewed runs** · **Joint combination not tested**

Rankings, LUCI reference, comparison, and coverage remain visible and usable.

Preparing a source-grounded explanation…

The explanation does not block the deterministic research results.

02 / Explanation modes

AI-assisted · citations validated

When live_ai is returned, show the source-grounded findings, applicability limits, missing evidence, and resolved citations. Never present unvalidated model output.

Reviewed fallback · provider unavailable

The same three explanation sections remain available using reviewed source-bound synthesis. Explain the reason without presenting it as a loss of scientific results.

Reason variants: provider not configured · timeout · provider failure · rate limit · citation validation failed

Example status: reviewed_fallback / not_configured / provider_key_absent

Evidence-gap safety warnings remain visible in both modes.

03 / Browse mode · no research question

Browse the research collection

Explore five investigation profiles, two reviewed candidate runs, and four phase observations. Add a question or choose a preset to evaluate a scenario.

Choose a scenario

Blank question is a valid browse state—not a request error. Unknown or unspecified conditions remain explicit.

04 / No reviewed runs match the current filters

No matching reviewed runs

This does not mean the material is safe. Broaden the filters or inspect published references; distinguish those references from reviewed run data.

Reset filters

The “no jointly tested combination” warning is a separate result state, not an empty catalog.

05 / Recovery & selection validation

Evaluation could not be loaded

Keep the scenario inputs intact. Do not label a previous evaluation as the result of the new scenario.

Retry evaluation

Explanation unavailable

Keep the completed evaluation visible. If the explanation request itself fails, offer retry; if the service returns reviewed fallback, render that explanation immediately.

Retry explanation

Select exactly two distinct reviewed runs to compare. One selected: “Choose one more run.” Selecting the same run twice is not a valid two-run comparison.

Download unavailable · keep the selected export visible and offer Retry download.

Integration contract

Evaluate first → render deterministic results → request explanation without blocking

POST /api/evaluate · question + scenario + selected_run_ids. Use resolved_scenario as the displayed source of truth. Alias run IDs normalize to BASSII_O15A / BASSII_O15B.

POST /api/explain · same question, scenario, and selected runs; include the evaluated catalog_version. Bind observed_findings, applicability_limits, missing_evidence, evidence_gap_warning, and citations_resolved.

GET /api/health · status, catalog_version, sha256_hash, counts, ai_provider_configured. The supplied document is a snapshot, not a live health result.

GET /api/export/runs · runs.csv | GET /api/export/phases · phases.csv

GET /api/export/dictionary · data_dictionary.json | GET /api/export/manifest · manifest.json

Retain X-Data-Version and X-Row-Count from CSV responses. Use the manifest for per-export integrity and count checks.

URL state · parse and serialize canonical scenario settings; strip secrets and prompts. Restore preset, parameters, and selected run IDs on reload or shared-link open.

Performance guidance · show evaluation immediately; provider requests have a 5.5 s timeout with automatic reviewed fallback. Do not imply a guaranteed end-to-end response time.